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MOST FREQUENTLY REPEATED CONCEPTUAL AREAS IN BIO506 - BIOCHEMISTRY II CONCEPT CRACKER MCQS FINAL EXAMS

Main Group (BS Biological Sciences) *https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN*

BIO506 – Biochemistry II *https://chat.whatsapp.com/LUxLhBNVSy3CYRTLXTapD1*

Q1: Glycosaminoglycans (GAGs) are composed of repeating _____ units.

Answer: Disaccharide

Q2: The amino sugar commonly found in GAGs is _____.

Answer: D-glucosamine

Q3: The acidic sugar that contributes a negative charge to GAGs is _____.

Answer: D-glucuronic acid

Q4: The extracellular matrix contains collagen, elastin, and _____.

Answer: Glycosaminoglycans (GAGs)

Q5: The high water-binding capacity of GAGs forms the body's _____ substance.

Answer: Ground

Q6: The only GAG that is neither sulfated nor attached to a protein is _____.

Answer: Hyaluronic acid

Q7: GAGs attached to proteins form molecules known as _____.

Answer: Proteoglycans

Q8: Proteoglycans are major structural components of the _____ matrix.

Answer: Extracellular

Q9: GAG chains are covalently attached to a _____ protein.

Answer: Core

Q10: GAG biosynthesis primarily occurs in the _____ apparatus.

Answer: Golgi

Q11: Glycosyltransferases catalyze the _____ of GAG chains.

Answer: Elongation

Q12: Amino sugars required for GAG synthesis are produced from _____-6-phosphate.

Answer: Fructose

Q13: The activated donor used during GAG synthesis is _____ derivatives.

Answer: UDP

Q14: UDP-GlcNAc stands for UDP-_____.

Answer: N-acetylglucosamine

Q15: Approximately 20% of glucose conversion in connective tissue is directed toward _____ synthesis.

Answer: Amino sugar

Q16: The precursor of glucuronic acid is _____-6-phosphate.

Answer: Glucose

Q17: Active glucuronic acid is donated as _____-glucuronic acid.

Answer: UDP

Q18: L-iduronic acid is produced by the enzyme _____.

Answer: Uronosyl-5-epimerase

Q19: The core protein of proteoglycans is synthesized on ribosomes of the _____ ER.

Answer: Rough

Q20: Initial folding of proteoglycan core proteins occurs inside the _____ lumen.

Answer: Endoplasmic reticulum

Q21: Glycosylation of proteoglycans mainly occurs in the _____ apparatus.

Answer: Golgi

Q22: The first sugar attached to the core protein during GAG synthesis is _____.

Answer: Xylose

Q23: Sulfation of GAG chains requires the sulfur donor _____.

Answer: PAPS

Q24: Sulfate groups are transferred by _____ enzymes.

Answer: Sulfotransferase

Q25: GAG degradation occurs inside the cellular _____.

Answer: Lysosomes

Q26: Lysosomal acid hydrolases function optimally at a pH of about _____.

Answer: 5

Q27: Extracellular GAG degradation begins with cellular _____.

Answer: Phagocytosis

Q28: The first enzymes involved in GAG degradation are _____.

Answer: Endoglycosidases

Q29: Sequential removal of terminal sugars is performed by _____.

Answer: Exoglycosidases

Q30: Defective GAG degradation results in _____ storage disorders.

Answer: Lysosomal

Q31: Hurler syndrome results from defective _____ degradation.

Answer: Glycosaminoglycan

Q32: Hunter syndrome follows an _____-linked inheritance pattern.

Answer: X

Q33: Glycoproteins contain relatively _____ carbohydrate chains.

Answer: Short

Q34: Glycoprotein carbohydrate chains usually contain _____ to 10 sugar units.

Answer: 2

Q35: N-linked glycoproteins are attached to the amino acid _____.

Answer: Asparagine

Q36: O-linked glycoproteins are attached to _____ or threonine residues.

Answer: Serine

Q37: ABO blood group antigens are examples of _____-linked glycoproteins.

Answer: O

Q38: High-mannose glycoproteins contain abundant _____ residues.

Answer: Mannose

Q39: Glycoprotein biosynthesis begins in the _____ endoplasmic reticulum.

Answer: Rough

Q40: O-linked glycosylation starts with the addition of _____.

Answer: N-acetylgalactosamine (GalNAc)

Q41: Membrane-bound glycosyltransferases are primarily located in the _____ apparatus.

Answer: Golgi

Q42: Glycoproteins destined for lysosomes contain _____-6-phosphate.

Answer: Mannose

Q43: Mannose-6-phosphate receptors are located in the _____ apparatus.

Answer: Golgi

Q44: The lipid carrier used in N-linked glycosylation is _____.

Answer: Dolichol

Q45: I-cell disease results from defective targeting of lysosomal _____.

Answer: Enzymes

Q46: Glycoprotein degradation follows the "_____ on, first off" principle.

Answer: Last

Q47: Misfolded glycoproteins in the ER are degraded through the _____ system.

Answer: Proteasomal

Q48: α-Mannosidosis is caused by deficiency of the enzyme _____.

Answer: α-Mannosidase

Q49: Mucopolysaccharidoses are caused by defects in _____ degradation.

Answer: Glycosaminoglycan

Q50: The most effective strategy for improving the prognosis of inherited lysosomal storage disorders is _____ diagnosis.

Answer: Early

Q51: Photosynthesis converts _____ energy into chemical energy.

Answer: Light

Q52: The organelle responsible for photosynthesis is the _____.

Answer: Chloroplast

Q53: The fluid-filled region inside chloroplasts is called the _____.

Answer: Stroma

Q54: The membrane structures where light reactions occur are the _____ membranes.

Answer: Thylakoid

Q55: Stacks of thylakoids are known as _____.

Answer: Grana

Q56: The primary photosynthetic pigment is _____.

Answer: Chlorophyll

Q57: Chlorophyll mainly absorbs _____ and red light.

Answer: Blue-violet

Q58: The accessory pigment that broadens light absorption is _____.

Answer: Carotenoid

Q59: The light-independent reactions are also known as the _____ cycle.

Answer: Calvin

Q60: The energy required for the Calvin cycle is produced during the _____ reactions.

Answer: Light-dependent

Q61: Photophosphorylation is the synthesis of _____ using light energy.

Answer: ATP

Q62: Photophosphorylation occurs on the _____ membrane.

Answer: Thylakoid

Q63: ATP synthesis during photosynthesis requires a _____ gradient.

Answer: Proton

Q64: Cyclic photophosphorylation involves only Photosystem _____.

Answer: I

Q65: Non-cyclic photophosphorylation requires Photosystem I and Photosystem _____.

Answer: II

Q66: The ultimate source of electrons in non-cyclic photophosphorylation is _____.

Answer: Water

Q67: Splitting of water during photosynthesis is called _____.

Answer: Photolysis

Q68: Photolysis releases oxygen, electrons, and _____.

Answer: Protons

Q69: The oxygen released during photosynthesis originates from _____.

Answer: Water

Q70: The final electron acceptor in the light reactions is _____.

Answer: $NADP^+$

Q71: Reduction of $NADP^+$ produces _____.

Answer: NADPH

Q72: ATP synthase synthesizes ATP through _____ phosphorylation.

Answer: Chemiosmotic

Q73: Photosystem II contains the reaction center pigment _____.

Answer: P680

Q74: Photosystem I contains the reaction center pigment _____.

Answer: P700

Q75: The electron transport chain is located within the _____ membrane.

Answer: Thylakoid

Q76: The movement of electrons generates a _____ motive force.

Answer: Proton

Q77: Cyclic photophosphorylation produces ATP but not _____.

Answer: NADPH

Q78: Non-cyclic photophosphorylation produces ATP, NADPH, and _____.

Answer: Oxygen

Q79: Carbon dioxide enters the Calvin cycle during the _____ phase.

Answer: Carbon fixation

Q80: The enzyme responsible for CO_2 fixation is _____.

Answer: RuBisCO

Q81: The first stable product of the Calvin cycle is _____.

Answer: 3-phosphoglycerate (3-PGA)

Q82: The carbon acceptor molecule in the Calvin cycle is _____.

Answer: Ribulose-1,5-bisphosphate (RuBP)

Q83: RuBisCO catalyzes the carboxylation of _____.

Answer: RuBP

Q84: ATP and NADPH are consumed during the _____ phase of the Calvin cycle.

Answer: Reduction

Q85: Regeneration of RuBP requires _____.

Answer: ATP

Q86: Three molecules of CO_2 produce one molecule of _____.

Answer: G3P

Q87: Glyceraldehyde-3-phosphate is abbreviated as _____.

Answer: G3P

Q88: Most G3P molecules are recycled to regenerate _____.

Answer: RuBP

Q89: Photosynthesis primarily occurs in the _____ cells of leaves.

Answer: Mesophyll

Q90: Gas exchange in leaves occurs through the _____.

Answer: Stomata

Q91: Guard cells regulate the opening and closing of _____.

Answer: Stomata

Q92: The major function of chlorophyll is to absorb _____ energy.

Answer: Light

Q93: Photosynthesis provides the primary source of _____ for nearly all living organisms.

Answer: Energy

Q94: Organisms capable of synthesizing their own food are called _____.

Answer: Autotrophs

Q95: Animals are classified as _____ because they cannot synthesize their own food.

Answer: Heterotrophs

Q96: The ATP produced during light reactions is mainly utilized in the _____ cycle.

Answer: Calvin

Q97: Chlorophyll molecules are embedded within the _____ membranes.

Answer: Thylakoid

Q98: The ultimate source of atmospheric oxygen is the process of water _____.

Answer: Photolysis

Q99: Light-independent reactions do not directly require _____.

Answer: Light

Q100: The process responsible for sustaining almost all life on Earth is _____.

Answer: Photosynthesis

Q101: The process in which ATP is synthesized using light energy is called _____.

Answer: Photophosphorylation

Q102: During non-cyclic photophosphorylation, electrons move from water to _____.

Answer: $NADP^+$

Q103: The reaction center of Photosystem II is designated as _____.

Answer: P680

Q104: The reaction center of Photosystem I is designated as _____.

Answer: P700

Q105: Water splitting occurs in association with Photosystem _____.

Answer: II

Q106: The enzyme responsible for ATP synthesis in chloroplasts is _____.

Answer: ATP synthase

Q107: The accumulation of protons inside the thylakoid lumen creates a _____ gradient.

Answer: Proton

Q108: The movement of protons through ATP synthase drives _____ synthesis.

Answer: ATP

Q109: Cyclic photophosphorylation returns electrons back to Photosystem _____.

Answer: I

Q110: The main advantage of cyclic photophosphorylation is additional production of _____.

Answer: ATP

Q111: Oxygen evolution occurs due to the _____ of water.

Answer: Photolysis

Q112: The oxygen-evolving complex is associated with Photosystem _____.

Answer: II

Q113: NADPH produced during light reactions provides _____ power for the Calvin cycle.

Answer: Reducing

Q114: ATP generated in light reactions supplies _____ for carbon fixation.

Answer: Energy

Q115: Carbon fixation begins with the enzyme _____.

Answer: RuBisCO

Q116: RuBisCO catalyzes the addition of CO_2 to _____.

Answer: RuBP

Q117: The first stable product after CO_2 fixation contains _____ carbon atoms.

Answer: Three

Q118: The reduction phase of the Calvin cycle converts 3-PGA into _____.

Answer: G3P

Q119: G3P is the precursor for the synthesis of _____.

Answer: Glucose

Q120: Most of the G3P produced is utilized for _____ of RuBP.

Answer: Regeneration

Q121: The Calvin cycle takes place in the chloroplast _____.

Answer: Stroma

Q122: Light reactions occur within the _____ membranes.

Answer: Thylakoid

Q123: Chlorophyll molecules are organized into _____.

Answer: Photosystems

Q124: Accessory pigments transfer absorbed energy to _____.

Answer: Chlorophyll a

Q125: Carotenoids mainly protect chloroplasts from _____ damage.

Answer: Photooxidative

Q126: The electron transport chain converts light energy into a _____ gradient.

Answer: Proton

Q127: Chemiosmosis couples proton movement with _____ synthesis.

Answer: ATP

Q128: ATP and NADPH together are required for _____ fixation.

Answer: Carbon

Q129: Photosynthesis stores solar energy as _____ energy.

Answer: Chemical

Q130: The major carbohydrate initially produced by photosynthesis is _____.

Answer: G3P

Q131: Six molecules of carbon dioxide are required to synthesize one molecule of _____.

Answer: Glucose

Q132: Chlorophyll absorbs light most efficiently in the red and _____ regions.

Answer: Blue-violet

Q133: The green appearance of leaves is due to _____ of green light.

Answer: Reflection

Q134: The Calvin cycle is also known as the _____ reduction cycle.

Answer: Carbon

Q135: ATP production during photosynthesis depends on _____ transport.

Answer: Electron

Q136: The primary electron donor replacing lost electrons in Photosystem II is _____.

Answer: Water

Q137: The light-dependent reactions directly generate ATP and _____.

Answer: NADPH

Q138: The Calvin cycle directly consumes ATP, NADPH, and _____.

Answer: CO_2

Q139: RuBP contains _____ carbon atoms.

Answer: Five

Q140: The enzyme RuBisCO is the most abundant _____ on Earth.

Answer: Enzyme

Q141: The primary role of stomata is regulating gaseous _____.

Answer: Exchange

Q142: Guard cells regulate water loss by controlling _____ opening.

Answer: Stomatal

Q143: The energy stored in ATP originates ultimately from _____.

Answer: Sunlight

Q144: Chloroplasts are found mainly in leaf _____ cells.

Answer: Mesophyll

Q145: The ultimate source of carbon in glucose is atmospheric _____.

Answer: Carbon dioxide

Q146: Photosynthesis reduces atmospheric CO_2 while releasing _____.

Answer: Oxygen

Q147: Plants capable of producing their own food are classified as _____.

Answer: Autotrophs

Q148: Organisms depending directly or indirectly on plants are called _____.

Answer: Heterotrophs

Q149: Photosynthesis is essential because it converts solar energy into _____ energy.

Answer: Chemical

Q150: The overall process that sustains nearly all ecosystems on Earth is _____.

Answer: Photosynthesis

Q151: The dietary source of galactose is primarily _____.

Answer: Lactose

Q152: Lactose is hydrolyzed by the enzyme _____.

Answer: Lactase (β-galactosidase)

Q153: Galactose is phosphorylated by the enzyme _____.

Answer: Galactokinase

Q154: Galactokinase converts galactose into _____.

Answer: Galactose-1-phosphate

Q155: The phosphate donor used by galactokinase is _____.

Answer: ATP

Q156: Galactose metabolism is closely integrated with _____ metabolism.

Answer: Glucose

Q157: Galactose ultimately enters glycolysis after conversion into _____.

Answer: Glucose-1-phosphate

Q158: UDP-galactose is converted into UDP-glucose by _____.

Answer: UDP-hexose 4-epimerase

Q159: UDP-galactose serves as a donor of galactose during _____ synthesis.

Answer: Lactose

Q160: UDP-galactose is also required for synthesis of _____ and glycolipids.

Answer: Glycoproteins

Q161: Even without dietary galactose, UDP-galactose can be synthesized from _____.

Answer: UDP-glucose

Q162: Deficiency of UDP-hexose 4-epimerase primarily affects _____ production.

Answer: UDP-glucose

Q163: Lactose is synthesized in the _____ apparatus.

Answer: Golgi

Q164: Lactose synthase consists of Protein A and _____.

Answer: Protein B

Q165: Protein A of lactose synthase is a _____.

Answer: β-D-galactosyltransferase

Q166: Protein B of lactose synthase is _____.

Answer: α-Lactalbumin

Q167: α-Lactalbumin is produced only in the _____ mammary gland.

Answer: Lactating

Q168: The hormone regulating α-lactalbumin synthesis is _____.

Answer: Prolactin

Q169: Hereditary fructose intolerance results from deficiency of _____.

Answer: Aldolase B

Q170: In hereditary fructose intolerance, _____ accumulates.

Answer: Fructose-1-phosphate

Q171: Hereditary fructose intolerance commonly develops after introduction of _____ into the diet.

Answer: Sucrose

Q172: Treatment of hereditary fructose intolerance requires elimination of fructose and _____.

Answer: Sucrose

Q173: Classic galactosemia results from deficiency of _____.

Answer: GALT

Q174: GALT stands for galactose-1-phosphate _____.

Answer: Uridylyltransferase

Q175: In classic galactosemia, galactose-1-phosphate and _____ accumulate.

Answer: Galactose

Q176: Galactosemia is managed by removing galactose and _____ from the diet.

Answer: Lactose

Q177: Galactokinase deficiency commonly causes _____.

Answer: Cataracts

Q178: Lactose intolerance is caused by deficiency of _____.

Answer: Lactase

Q179: Symptoms of lactose intolerance usually appear within _____ hours after lactose ingestion.

Answer: Two

Q180: The pentose phosphate pathway primarily produces NADPH and _____ phosphate.

Answer: Ribose

Q181: The rate-limiting enzyme of the pentose phosphate pathway is _____.

Answer: Glucose-6-phosphate dehydrogenase (G6PD)

Q182: G6PD catalyzes oxidation of _____.

Answer: Glucose-6-phosphate

Q183: The oxidative phase of PPP generates _____.

Answer: NADPH

Q184: The major sugar produced during the oxidative phase is _____.

Answer: Ribulose-5-phosphate

Q185: NADPH is mainly required for _____ biosynthesis.

Answer: Reductive

Q186: High intracellular NADPH _____ G6PD activity.

Answer: Inhibits

Q187: Increased oxidative stress _____ PPP activity.

Answer: Increases

Q188: The non-oxidative phase of PPP produces ribose-5-phosphate and glycolytic _____.

Answer: Intermediates

Q189: The enzyme transferring two-carbon units in PPP is _____.

Answer: Transketolase

Q190: The enzyme transferring three-carbon units in PPP is _____.

Answer: Transaldolase

Q191: The non-oxidative reactions of PPP are _____.

Answer: Reversible

Q192: Ribose-5-phosphate is required for _____ synthesis.

Answer: Nucleotide

Q193: G6PD deficiency is inherited as an _____-linked disorder.

Answer: X

Q194: G6PD deficiency primarily affects _____ blood cells.

Answer: Red

Q195: Heinz bodies are composed of denatured _____.

Answer: Hemoglobin

Q196: Favism is triggered by consumption of _____ beans.

Answer: Fava

Q197: Reduced NADPH production increases susceptibility to _____ stress.

Answer: Oxidative

Q198: A major antioxidant therapy for G6PD deficiency includes _____ E.

Answer: Vitamin

Q199: CRISPR-Cas9 is being explored as a _____ therapy approach for G6PD deficiency.

Answer: Gene

Q200: The major clinical consequence of severe G6PD deficiency is _____ anemia.

Answer: Hemolytic

Q201: Glycosaminoglycans (GAGs) are _____ heteropolysaccharides.

Answer: Unbranched

Q202: GAGs consist of repeating _____ units.

Answer: Disaccharide

Q203: The amino sugar commonly present in GAGs is _____.

Answer: D-glucosamine

Q204: The acidic sugar commonly present in GAGs is _____.

Answer: D-glucuronic acid

Q205: The negative charge of GAGs is mainly due to _____ and sulfate groups.

Answer: Carboxyl

Q206: The only GAG that is not sulfated is _____.

Answer: Hyaluronic acid

Q207: GAGs attached to proteins form _____.

Answer: Proteoglycans

Q208: The major component of proteoglycans is the _____ protein.

Answer: Core

Q209: GAG biosynthesis mainly occurs in the _____ apparatus.

Answer: Golgi

Q210: The enzyme responsible for elongation of GAG chains is _____.

Answer: Glycosyltransferase

Q211: UDP-GlcNAc is synthesized from _____-6-phosphate.

Answer: Fructose

Q212: The first sugar attached to the core protein is _____.

Answer: Xylose

Q213: Sulfation of GAG chains requires the sulfur donor _____.

Answer: PAPS

Q214: Sulfate groups are transferred by _____ enzymes.

Answer: Sulfotransferases

Q215: GAG degradation occurs mainly inside _____.

Answer: Lysosomes

Q216: Acid hydrolases function optimally at a pH of about _____.

Answer: 5

Q217: GAG degradation begins with _____.

Answer: Endoglycosidases

Q218: Terminal sugars are removed by _____.

Answer: Exoglycosidases

Q219: Defective degradation of GAGs leads to _____ storage disorders.

Answer: Lysosomal

Q220: Hunter syndrome follows an _____-linked inheritance pattern.

Answer: X

Q221: Glycoproteins contain relatively _____ carbohydrate chains.

Answer: Short

Q222: N-linked glycoproteins are attached to the amino acid _____.

Answer: Asparagine

Q223: O-linked glycoproteins are attached to _____ residues.

Answer: Serine

Q224: The first sugar added during O-linked glycosylation is _____.

Answer: N-acetylgalactosamine (GalNAc)

Q225: N-linked glycoproteins are synthesized on the lipid carrier _____.

Answer: Dolichol

Q226: Glycoprotein biosynthesis begins in the _____ endoplasmic reticulum.

Answer: Rough

Q227: Final processing of glycoproteins occurs in the _____ apparatus.

Answer: Golgi

Q228: Glycoproteins destined for lysosomes possess _____-6-phosphate residues.

Answer: Mannose

Q229: Mannose-6-phosphate receptors are located in the _____ apparatus.

Answer: Golgi

Q230: I-cell disease results from defective _____ of lysosomal enzymes.

Answer: Targeting

Q231: Misfolded glycoproteins are degraded through the _____ system.

Answer: Proteasomal

Q232: Glycoprotein degradation follows the "_____ on, first off" principle.

Answer: Last

Q233: α-Mannosidosis results from deficiency of _____.

Answer: α-Mannosidase

Q234: Glycoprotein storage diseases are usually inherited as _____ recessive disorders.

Answer: Autosomal

Q235: Mucopolysaccharidoses result from defective degradation of _____.

Answer: Glycosaminoglycans

Q236: The incidence of mucopolysaccharidoses is approximately one in _____ live births.

Answer: 25,000

Q237: Most mucopolysaccharidoses are inherited as _____ recessive disorders.

Answer: Autosomal

Q238: Diagnosis of mucopolysaccharidoses commonly involves _____ analysis.

Answer: Urine

Q239: Enzyme activity is confirmed by _____ assays.

Answer: Biochemical

Q240: Bone marrow transplantation introduces functional _____.

Answer: Macrophages

Q241: Enzyme replacement therapy cannot effectively reverse _____ damage.

Answer: Neurological

Q242: Glycoprotein metabolism disorders primarily result from deficiency of _____ enzymes.

Answer: Lysosomal

Q243: Sialidosis is caused by deficiency of the enzyme _____.

Answer: Neuraminidase

Q244: I-cell disease is also known as mucolipidosis type _____.

Answer: II

Q245: Deficiency of α-mannosidase causes _____.

Answer: α-Mannosidosis

Q246: Diagnosis of glycoprotein storage disorders may require _____ testing.

Answer: Genetic

Q247: Stem cell transplantation may provide functional _____ enzymes.

Answer: Lysosomal

Q248: Early diagnosis helps _____ disease progression.

Answer: Slow

Q249: Supportive management includes physical, occupational, and _____ therapy.

Answer: Speech

Q250: Most inherited glycoprotein metabolism disorders currently have no definitive _____.

Answer: Cure

Q251: Photosynthesis converts light energy into _____ energy.

Answer: Chemical

Q252: The light-dependent reactions occur in the _____ membrane.

Answer: Thylakoid

Q253: The Calvin cycle takes place in the chloroplast _____.

Answer: Stroma

Q254: The primary pigment responsible for photosynthesis is _____.

Answer: Chlorophyll a

Q255: The accessory pigments mainly function to _____ the absorption spectrum.

Answer: Broaden

Q256: Photosystem II has a reaction center known as _____.

Answer: P680

Q257: Photosystem I has a reaction center known as _____.

Answer: P700

Q258: Water is split during the process called _____.

Answer: Photolysis

Q259: The oxygen released during photosynthesis originates from _____.

Answer: Water

Q260: The final electron acceptor of the light reactions is _____.

Answer: $NADP^+$

Q261: Reduction of $NADP^+$ produces _____.

Answer: NADPH

Q262: ATP synthesis in chloroplasts is driven by a _____ gradient.

Answer: Proton

Q263: The movement of electrons through the electron transport chain creates a _____ motive force.

Answer: Proton

Q264: Cyclic photophosphorylation produces ATP but not _____.

Answer: NADPH

Q265: Non-cyclic photophosphorylation produces ATP, NADPH, and _____.

Answer: Oxygen

Q266: The enzyme responsible for carbon fixation is _____.

Answer: RuBisCO

Q267: The carbon acceptor molecule in the Calvin cycle is _____.

Answer: Ribulose-1,5-bisphosphate (RuBP)

Q268: The first stable product of carbon fixation is _____.

Answer: 3-Phosphoglycerate (3-PGA)

Q269: ATP and NADPH are mainly consumed during the _____ phase of the Calvin cycle.

Answer: Reduction

Q270: Most glyceraldehyde-3-phosphate is used for regeneration of _____.

Answer: RuBP

Q271: Six molecules of carbon dioxide are required to synthesize one molecule of _____.

Answer: Glucose

Q272: Chlorophyll reflects _____ light.

Answer: Green

Q273: Stomata regulate gaseous _____ in leaves.

Answer: Exchange

Q274: Guard cells regulate the opening and closing of _____.

Answer: Stomata

Q275: The ATP produced during light reactions is mainly consumed in the _____ cycle.

Answer: Calvin

Q276: Ketone bodies are synthesized primarily in the _____.

Answer: Liver

Q277: Ketogenesis occurs in the _____ of liver cells.

Answer: Mitochondria

Q278: The major precursor for ketone body synthesis is _____.

Answer: Acetyl-CoA

Q279: The first ketone body formed during ketogenesis is _____.

Answer: Acetoacetate

Q280: Acetoacetate can be reduced to _____.

Answer: β-Hydroxybutyrate

Q281: Acetoacetate may spontaneously decarboxylate to form _____.

Answer: Acetone

Q282: The rate-limiting enzyme of ketogenesis is _____ synthase.

Answer: HMG-CoA

Q283: Ketogenesis increases during prolonged _____.

Answer: Fasting

Q284: Uncontrolled _____ mellitus is a common cause of excessive ketogenesis.

Answer: Diabetes

Q285: Extrahepatic tissues utilize ketone bodies through the process of _____.

Answer: Ketolysis

Q286: The liver cannot utilize ketone bodies because it lacks _____.

Answer: Thiophorase

Q287: Thiophorase is also known as succinyl-CoA:acetoacetate _____.

Answer: CoA transferase

Q288: The brain increasingly utilizes ketone bodies during prolonged _____.

Answer: Starvation

Q289: Excessive accumulation of ketone bodies causes metabolic _____.

Answer: Acidosis

Q290: Diabetic ketoacidosis is characterized by elevated blood _____ bodies.

Answer: Ketone

Q291: Fruity odor of breath in diabetic ketoacidosis is caused by _____.

Answer: Acetone

Q292: Urinary ketone bodies are detected by the _____ test.

Answer: Nitroprusside

Q293: The normal physiological role of ketone bodies is to serve as an alternative _____ source.

Answer: Energy

Q294: During prolonged fasting, ketone bodies reduce the requirement for _____.

Answer: Glucose

Q295: Increased fatty acid β-oxidation provides excess _____ for ketogenesis.

Answer: Acetyl-CoA

Q296: Oxaloacetate depletion during fasting favors _____ body formation.

Answer: Ketone

Q297: The major circulating ketone body is _____.

Answer: β-Hydroxybutyrate

Q298: Severe ketoacidosis lowers blood _____.

Answer: pH

Q299: Prompt administration of insulin suppresses excessive _____ in diabetic ketoacidosis.

Answer: Ketogenesis

Q300: Disorders of ketone body metabolism primarily affect the body's _____ balance.

Answer: Energy

🚨 Concept Cracker MCQS Most Important Repeated Conceptual 🚨

⚠ Important Note:

These MCQs have been compiled from repeated exam patterns, important concepts, and previous assessments to support your preparation.

📚 However, do NOT rely solely on this compilation.

For the best results, thoroughly study your Handouts, PPTs, Video Lectures, and Course Material as well.

Prepare Smart, Work Hard, and Aim for Excellence.

Best of Luck! 🍀📚

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